

HARSHAL KOLHE

 [GitHub](#) |  [LinkedIn](#) |  harshalkolhe03@gmail.com

EDUCATION

Visvesvaraya National Institute of Technology, India

Bachelor of Technology in Chemical Engineering, Minor in Internet of Things

Aug' 23 – May' 27

CGPA: 8.66/10.0

EXPERIENCE

Indian Institute of Technology, Roorkee

May' 26 – Jul' 26

Summer Research Intern | [\[Certificate\]](#)

- Designed a custom secure drone authentication protocol leveraging V2X device communications to enable secure inter-drone communication and authentication.
- Implemented a low-latency communication protocol in Embedded C on the Unex SOM-352 chipset, implementing IEEE 802.11p standards to enable secure authentication and data sharing with 10 ms communication latency.
- Validated the end-to-end communication and authentication system onboard two self-assembled UAVs, demonstrating a successful real-world proof of concept for secure low-latency inter-UAV communication.

PROJECTS

Dual UAV Scan and Spray Architecture for Precision Agriculture | [\[GitHub\]](#)

Jan' 26 – Apr' 26

- Developed a real-time onboard crop disease detection pipeline using OpenCV and HSV-based image segmentation, under varying outdoor lighting conditions through dual-mask filtering and ND-filter-assisted preprocessing.
- Built a dual-nozzle spraying system with real-time nozzle diameter selection based on detected plant clusters; developed the Raspberry Pi-based control algorithm and designed, fabricated, and assembled the control PCB for automated operation.
- Designed and fabricated a custom carbon-fiber quadcopter frame capable of carrying a 10 kg fluid payload.

Multiagent Hybrid Disaster Rescue Drones | [\[GitHub\]](#)

Jul' 25 – Jan' 26

- Developed a two-drone system that searches in a lawnmower pattern, detects humans, and diverts to deliver relief packages.
- Fine-tuned and deployed a YOLO object detection model for aerial imagery, achieving 90% human detection accuracy at 8m, and integrated it into an end-to-end autonomous localisation pipeline that generated target GPS coordinates and performed dynamic UAV redirection with cluster-based filtering.
- Developed the core telemetry processing layer for a multi-UAV visualisation platform, combining MAVLink data streams and enabling real-time tracking, status monitoring, and mission visualisation.

Design and Development of Fixed Wing UAV | [\[GitHub\]](#)

May' 24 – Oct' 24

- Constructed a custom fixed-wing UAV, with a 1.25m wingspan, 5:1 aspect ratio, and 25 cm chord for aerodynamic performance.
- Conducted aerodynamic analysis of various airfoils in XFLR5 to achieve a target 2D C_L of 1.0 at 4–5° angle of attack and made the aircraft's CAD model based on aircraft design principles.

PUBLICATIONS

- R. Deshmukh, **H. Kolhe**, S. Biswas, and S. S. Chiddarwar, "Hybrid Controller Development for a VTOL Using PID, LQR and MPC," presented at the International Conference on Advances in Mechanical and Manufacturing Systems (ICAMMS), VNIT Nagpur, India, 2026.
- S. Kshirsagar, **H. Kolhe**, R. Deshmukh, K. Ayush, S. Dhalwar, S. Biswas, M. Limaye, and S. S. Chiddarwar, "A Multi-Agent Drone System for Real-Time Victim Detection and Aid Delivery," presented at the International Conference on Advances in Mechanical and Manufacturing Systems (ICAMMS), VNIT Nagpur, India, 2026.
- A. Sharma, M. Ravikiran, S. Chakrabarty, R. Saluja, M. Limaye, **H. Kolhe**, and S. Samiron, "Prompted to Fly: Translating Free-Form Instructions into Schema-Constrained Mission Generation for UAVs using LLMs," 2025 IEEE International Conference on Big Data (BigData), Macau, China, 2025, pp. 7278-7286, doi: [10.1109/BigData66926.2025.11401432](https://doi.org/10.1109/BigData66926.2025.11401432).

SKILLS

Languages: C, C++, Python, Java

Software and Tools: ArduPilot, QGroundControl, Mission Planner, Linux, ROS2, MATLAB

Libraries: OpenCV, Ultralytics (YOLOv8), NumPy, Pandas, Matplotlib, DroneKit, MAVSDK, MAVlink

EXTRACURRICULAR ACTIVITIES

- Secured the [Best Paper Presentation Award](#) at the ICAMMS Conference for presenting research work on autonomous systems.
- Secured the Ignite Prize in **NIDAR**, a national-level UAV autonomy Competition, placing among the top 5 out of 100+ teams.
- Secured [First Runner-up](#) prize at the Drone Helix Competition, SVNIT Surat.
- Vice Chairman of IvLabs, Robotics and Artificial Intelligence Lab at VNIT, Nagpur.